

Autodesk® **Scaleform**®

Scaleform 4.2

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Autodesk® Scaleform® 4.2

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Scaleform 4.0

Scaleform SDK

Scaleform 4.0 API

Scaleform

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Scaleform 3.x

Scaleform 4.0

API

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Renderer2D

2. ž ¥ r > 4 _ • u i — - !

GFxLoader

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4. ž ¥ r œ ¥ b • 4 %

MovieDisplayHandle

Renderer2D::Display

5. ' 4 ... 4 - i • u z r X ¥ GFx::Movie

6.] k q ' ž ¥ r > 4

GRenderer

API

Render::HAL

Scaleform 3.x

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Scaleform
 Renderer2D
 Display
 Render::PS3::HAL
 Render::Renderer2D
 Render::HAL
 Renderer2D
 Render::D3D9::HAL

```
D3DPRESENT_PARAMETERS PresentParams;
IDirect3DDevice9* pDevice = ...;
ThreadId renderThreadId = Scaleform::GetCurrentThreadId();
Render::ThreadCommandQueue *commandQueue = ...;
Ptr< Render::D3D9::HAL> pHal;
Ptr< Render::Renderer2D> pRenderer;

pHal = *SF_NEW Render::D3D9::HAL( commandQueue);

if (!pHal->InitHAL(Render::D3D9::HALInitParams(pDevice, PresentParams, 0,
renderThreadId)))
{
return false;
}
pRenderer = * SF_NEW Render :: Renderer2D ( pHal );
```

PresentParams hWnd pDevice
 renderThreadId
 ThreadCommandQueue

Renderer2D HAL

2.2 ž ¥ r 4 _ • u i — - Ò

TextureHeight Scaleform 3.3 TextureWidth
 NumTextures
 MaxSlotHeight

 InitHAL
 InitHAL SetParams

2.3 x a k s • œ o 4 k - Ç

Scaleform 4.0 Gfx::ImageCreator
 Render::TextureManager Scaleform
 ImageCreator Gfx::Loader

```
Gfx::Loader      loader;
...
SF::Ptr<      Gfx::ImageCreator>      pimageCreator      =
*      new Gfx::ImageCreator(      pRenderThread-      >GetTextureManager());
loader . SetImageCreator      ( pimageCreator      );
```

ImageCreator

TextureManager

Render::HAL::GetTextureManager()

TextureManager
 GetTextureManager

CreateTexture

TextureManager

Render::HAL

x

D3DCREATE_MULTITHREADED

D3D9

x

Render::HAL

ThreadCommandQueue

D3D y, V k Q ž " % • • s k ž u { ž ¥ r œ ¥ b Q ·
* . x a k s • - Ç * ì % ThreadCommandQueue - ð Ž ' Ñ * â % - g 4 ... k
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Platform::RenderHALThread

2.4 ž ¥ r œ ¥ b • 4 ‰

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draw frame

Scaleform 4.0 API

GFx::MovieDisplayHandle

GFx::Movie

Display

GFx::Movie

1. B ò - GFx::MovieDef::CreateInstance

GFx::Movie

2. “V ¥ k ž u { ê ö • 4 ‰ -

Advance

ActionScript

Advance

Invoke/DirectAccess API

Advance

Scaleform

3. ž ¥ r œ ¥ b – MovieDisplayHandle

```
//-----  
// 1.  
  
Ptr<GFx::Movie>      pMovie = ...;  
GFx::MovieDisplayHandle hMovieDisplay;  
  
//  
//  
pMovie->SetViewport(width, height, 0,0, width, height);  
hMovieDisplay = pMovie->GetDisplayHandle();  
  
//  
// Scaleform Player  
pRenderThread->AddDisplayHandle(hMovieDisplay);  
  
//-----  
// 2.  
  
//                                     ExternalInterface  
//                                     Advance  
float deltaT = ...;  
pMovie->HandleEvent(...);  
pMovie->Advance(deltaT);  
  
//  
//  
pRenderThread->WaitForOutstandingDrawFrame();  
pRenderThread->DrawFrame();  
  
//-----  
// 3. - DrawFrame  
Ptr<Render::Renderer2D> pRenderer = ...;  
  
pRenderer->BeginFrame();  
bool hasData = hMovieDisplay.NextCapture(pRenderer->GetContextNotify());  
if (hasData)  
    pRenderer->Display(hMovieDisplay);  
pRenderer->EndFrame();
```

x WaitForOutstandingFrame – CPU 2

Advance

x MovieDisplay.NextCapture – Advance

false

2.5 ' 4 ... 4 - i • u z r X ¥

GFx::Movie Release

Render::ThreadCommandQueue

ThreadCommandQueue

Movie::ShutdownRendering(false)

Movie::IsShutdownRenderingComplete

NextCapture

false

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GfX::Movie::Advance

ActionScript

Scaleform

Render::HAL

Scaleform

x

x

Scaleform Player

Scaleform

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Advance

Scaleform

GFx::Movie::Advance
GFx::Movie::Capture

NextCapture

x Movie::Advance

x Capture NextCapture

Capture
Capture

x Invoke
Capture

Direct Access API
Advance

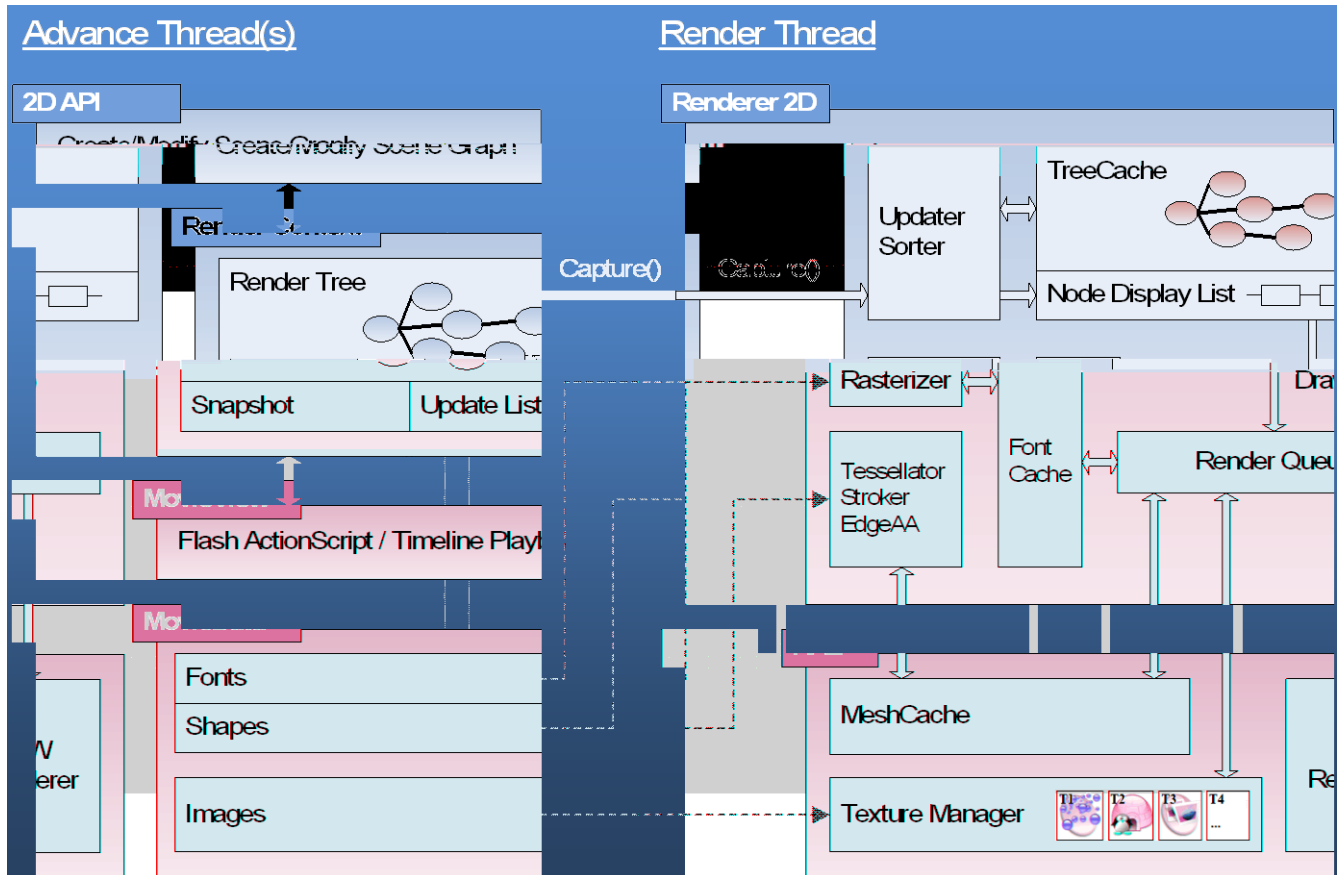
3.3 Scaleform 4.0
Movie::Invoke

Display
Capture Advance
Invoke

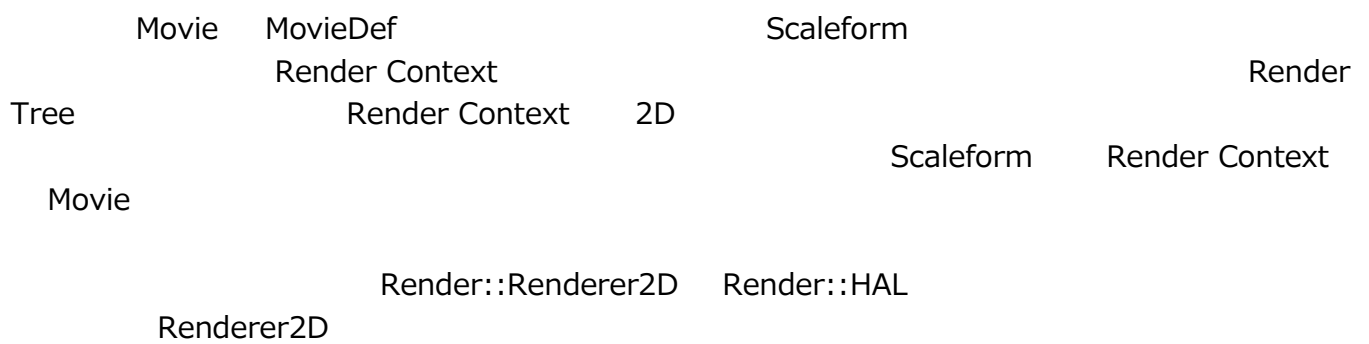
Scaleform
Advance

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Scaleform 4.0



1.



Display

Render::HAL HAL

Render HAL

x

x Font Cache

Font Cache

x

 Mesh Cache

x Render Queue

/

 CPU

x HAL API

ActionScript

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Scaleform

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ActionScript	AS2	rendererString	rendererFloat
AS3	setRendererString	setRendererFloat	
	AS2/AS3		string
float		UserDataState::Data	
HAL::PushUserData		Render::HAL's UserDataStack	

Position2dOffset

```
<ShaderSource id="Position2dOffset" pipeline="Vertex">
  Position2d(
    attribute float4 pos   :          POSITION,
    varying   float4 vpos  :          POSITION,
    uniform   float4 mvp[2],
    uniform   float poffset)
  {
    vpos = float4(0,0,0,1);
    vpos.x = dot(pos, mvp[0]) + poffset;
    vpos.y = dot(pos, mvp[1]) + poffset;
  }
</ShaderSource>
```

x y poffset

Position2d

D3D9_Shader.cpp

ShaderInterface::SetStaticShader:

```
// See if we have user data on the stack.
if (pHal->UserDataStack.GetSize() > 0 )
{
    const UserDataState::Data* data = pHal->UserDataStack.Back();
    if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
    data->RendererString.CompareNoCase("abc") == 0)
    {
        // Modify the shader we use, if we find the matching user data we were
        expecting.
        vshader = (VertexShaderDesc::ShaderType)((int)vshader +
VertexShaderDesc::VS_base_Position2dOffset);
        shader = (FragShaderDesc::ShaderType) ((int)shader +
FragShaderDesc::FS_base_Position2dOffset);
    }
}
```

HAL UserDataStack

abc

Position2dOffset

ShaderData.xml

D3D9_ShaderDescs.h

poffset
ShaderInterface::Finish:

```
// See if we have user data on the stack.
if (pHal->UserDataStack.GetSize() > 0 )
{
    const UserDataState::Data* data = pHal->UserDataStack.Back();
    if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
    data->RendererString.CompareNoCase("abc") == 0)
    {
        // Add the poffset uniform into the uniform data.
        for ( unsigned m = 0; m < Alg::Max<unsigned>(1, meshCount); ++m)
        {
            float poffset = data->RendererFloat / 256.0f;
            SetUniform(CurShaders, Uniform::SU_poffset, &poffset, 1, 0, m);
        }
    }
}
```

meshCount

meshCount
HAL

m

5.3] k q ' { Ÿ 4 - , u s Q î • * J

HAL Scaleform

Scaleform

Scaleform

AS2 "disableBatching"

AS3

"disableBatching"

AS2/AS3

6 GFxShaderMaker

6.1 ê

Scaleform 4.1 GFxShaderMaker Scaleform
Render_D3D9
ShaderData.xml

D3D9

Src/Render/D3D9_ShaderDescs.h
Src/Render/D3D9_ShaderDescs.cpp
Src/Render/D3D9_ShaderBinary.cpp

6.2 3 / i Y 4 r 4 API Q g • 4 z J HAL

6.2.1 D3D9

D3D9 HAL ShaderModel 2.0 ShaderModel 3.0
HAL

GFxShaderMaker '-shadermodel'

GFxShaderMaker.exe -platform D3D1x -shadermodel SM30

ShaderModel 2.0 ShaderModel 2.0
InitHAL

6.2.2 D3D1x

D3D1x HAL

3

D3D11 D3D10.1

FEATURE_LEVEL_10_0

FEATURE_LEVEL_9_3

FEATURE_LEVEL_9_1

HAL

GFxShaderMaker '-featurelevel'

GFxShaderMaker.exe -platform D3D1x - featurelevel

FEATURE_LEVEL_10_0,FEATURE_LEVEL_9_1

FEATURE_LEVEL9_3

FEATURE_LEVEL_9_3

InitHAL

FEATURE_LEVEL_9_1